



Putting Safety of Intended Functionality SOTIF into Practice

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Abstract

The increase of autonomy demand in the automotive industry made the usage of AI models inevitable. However, such models introduce a variety of threats to automobile safety and security. ISO/PAS 21448 SOTIF is a safety standard that is designed to deal with risks due to non-electrical and non-electronic failures. In this paper we put SOTIF into practice. In our work we introduce a conceivable safety critical scenario that targets the lane keep assist function. We use the suggested modelling techniques in the SOTIF standard to

analyze the scenario and extract the trigger event. In result, we propose a contextual based predictive ML model to monitor the intervention between the driver and lane keep assist system. Our approach followed the SOTIF verification and validation guidelines. Empirically, we use a real safety critical scenario dataset as well as an augmented dataset. Our results show a high precision/recall values that exceed 90% by an increase of more than 150% in f1 score compared to non contextual models. It also showed that there is a trade-off relation between the precision/recall values and the sensitivity of models to its inputs.

Introduction

Autonomy became one of the features that characterize the automotive industry trends [1]. With having automotive industry heading towards such transformation phase, the nature of the industry is changing from being a purely mechanical based system to a Cyber-physical system (CPS), where software is to be responsible for handling different components of a vehicle starting from the core functions like steering column, up to vehicle-to-vehicle communication and communication with other devices in roads in addition to vehicle luxuries.

This transformation introduces new challenges in terms of safety. Firstly, Due to the excellence of Artificial intelligence AI and machine learning ML models in solving complex problems, they became a major part in many autonomous applications. Their ability to provide high quality solutions using limited data with high generalization capabilities [2] made them a preferable candidate for developing autonomous functions. However AI models form a threat to automobile safety. Since these models are characterized by limited transparency, presence of non determinism and uncertainty about error rate [2-4]. Also their behavior as a black box hardens the validation and verification of such models. [5-7]. Moreover, various automation levels defined in [8] introduce the factor of intervention between humans and machines. Which could lead to unsafe situations when the user misuses a function as explained in [9]. Additionally, the absence of a safety standard that addresses such challenges.

In this work, we are specifically concerned with the development of safe AI based solutions. These solutions must satisfy the operational requirements such as accuracy, as well as the safe operation of the models in unusual scenarios.

To address these challenges a new safety standard was introduced. Namely, SOTIF Safety of the intended functionality. Unlike ISO 26262, SOTIF is concerned with safety issues due to non electrical/electronic failures. SOTIF provides guidelines for design, validation and verification processes in order to develop and evaluate the safety of autonomous systems.

In this paper we propose safe development of a contextual based ML model, that monitors the intervention between driver and lane keep assist system. We do this by putting SOTIF into practice, at first, we introduce a real safety critical scenario that targets the lane keep assist function. Secondly, we use the suggested modelling technique in SOTIF to analyze the scenario. Then we follow the guidelines of SOTIF through validation and verification in order to evaluate the safety of our model. At last we show performance of our models that score high precision/recall values that exceed 90% by an increase of more than 150% in f1 score compared to non contextual models.

The paper proceeds as follows: (1) We give an overview on the SOTIF standard. (2) We introduce our safety critical scenario. (3) We discuss related work from various domains. (4) We discuss the design of the proposed models and their operation. (5) We proceed with introducing the datasets and training configurations. (6) We explain the verification and validation test cases. (7) We introduce results and discuss the solutions.

Safety of the Intended Functionality SOTIF

Previously electrical and electronic failures formed the major threat for automobile safety. ISO 26262 was concerned with this type of failure. With the increase of autonomy and the wide adoption of ML based components, different sources of threat were introduced. Old methodologies in ISO 26262 could not deal with this type of threat, therefore the SOTIF standard was designed. SOTIF aims to provide guidelines for design, verification and validation processes. Verification is the set of activities to gain confidence of proper operation against known test cases, while validation is the set of activities to gain confidence of proper operation against unknown test cases [9]. It also suggests how to explore a potential risk area that covers most of the cases that an automobile could pass by. SOTIF introduced itself as a complementary document to ISO 26262. It is concerned with faults due to user misuse, limited performance of sensors or sudden changes in environment that could threaten user safety. SOTIF excludes threats due to Cyber attacks.

SOTIF emphasizes the importance of contextual analysis. It provides a modelling technique for events versus time illustrated in Figure 1.

Figure 1 illustrates the proposed method by SOTIF to mitigate a risk scenario. The risk scenario is broken into a set of scenes where each scene impacts the following one.

In Figure 1 **Goal** defines the objective or the intended functionality of a vehicle from a certain perspective. While the **Scene** is a description of the current state and the environment of a vehicle. A combination between the current state and the objective to be accomplished is called **Situation**. The driving function in Figure 1 can be an AI model that takes Actions depending on the current state for a certain objective.

The system moves forward where the scene is updated with a new event and the taken action. Then it is fed again with the objective for the driving function for an action and so on.

SOTIF suggests a technique to address a safety issue of a component that could briefly be described in the following steps: (1) Determine system specification, (2) Identify acceptable triggering events, (3) Evaluate the known hazardous

scenarios, (4) Evaluate hazardous unknown scenarios. These activities are done in an iterative way.

Through the paper we use this methodology to explain our safety critical scenario, that could cause harm to passengers or damage to the automobile.

Safety Critical Scenario

We demonstrate a safety critical scenario, which could occur due to user misuse of a certain functionality combined with limited sensor performance. For this we designed a scenario that targets the lane keep assistant function. In this scenario, trivial conceivable misuse could lead to serious damage.

Scenario Components

Our scenario targets the Lane Keep Assistant (LKA) which is a safety critical function designed to work in high speed roads. LKA works by receiving the position of a vehicle within a lane. The position can be provided from a camera sensor or other sensors. An Electronic Control Unit (ECU) is responsible for controlling the vehicle to keep it within the lane. LKA is a feature that is normally controlled by the user. It has two states of operation ON and OFF. However, even when the LKA is ON the user can override the decisions of the LKA by holding the steering wheel. LKA remains in its state during and after the user overrides. The following scenario is applied to an automobile that uses CAN protocol for internal communication.

- 1) Lane Keep Assist (LKA): The frame responsible for changing the LKA states has a specific ID and specific content. The operation of the LKA state works as a toggle button, such that if the current state is OFF and that specific frame is received the state is converted to ON and vice versa.
- 2) Right Lane Distance (RLD): RLD frame has a unique ID. It carries the data from the sensor that defines the distance between the vehicle and the right lane.

Normally, the RLD takes a limited range of values while moving straight forward in the lane. At turning left a larger range of values are observed, while at turning right smaller values are observed. Depending on RLD values and the state of LKA, the ECU will decide to modify the direction of the vehicle.

Scenario

Scenario illustrated in Figures 2 and 3 introduces a safety critical case that occurs due to a combination of user misuse and sensor limited performance.

Initially, we assume that the LKA was OFF. The Driver was controlling the vehicle without any assistance. Driver decided to switch lanes. Then, while being in a hurry. During switching lanes the driver decided to turn the LKA ON to let

FIGURE 1 Proposed contextual modelling technique in SOTIF document

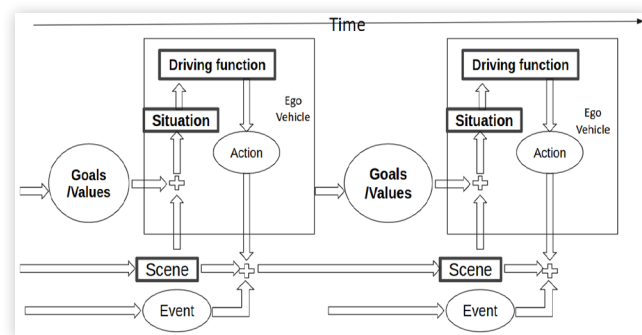


FIGURE 2 Scenario A, first scene. Using SOTIF standard modeling technique, the driver decided to turn LKA ON while switching lanes, he took his hands off the steering wheel when LKA was in a hesitation phase

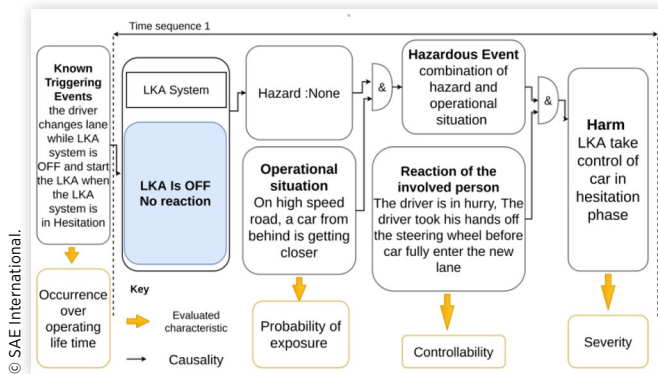
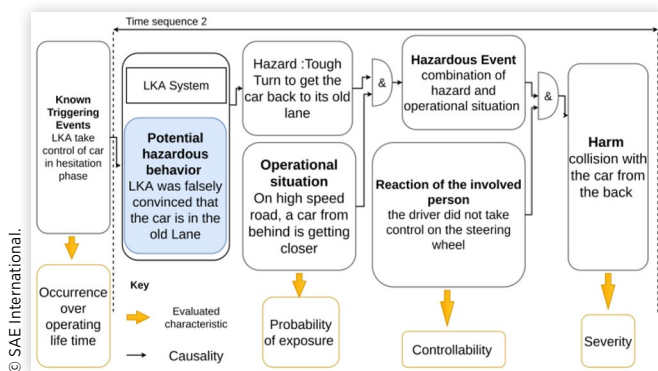


FIGURE 3 Scenario A, Second scene. The LKA took control of the vehicle after the user took his hands off the steering wheel. LKA is at a hesitation phase so it cannot decide accurately which lane the vehicle belongs to. LKA steers the vehicle to get it back to the old lane leading to a collision between a vehicle coming from behind. The driver could not handle the case due to shock



the LKA system control the steering function. Concurrently, the driver took hands off the steering wheel. This last decision represents the user misuse of the function. While the limited sensor performance is represented in the hesitation of the RLD values during switching lanes. Where the LKA system could not decide the current lane correctly. LKA was falsely convinced that the vehicle was still in the old lane on its way to leave the lane. Therefore a tough turn was made by the LKA to return the vehicle to its original lane.

Hesitation phase is where the sensors which calculate the RLD are uncertain about the reference line. The old lane right line or the new lane right line. It occurs in real scenarios when automobiles switch lanes.

In Figures 2 and 3 we show a scenario where the vehicle was moving on a high speed road. Our vehicle driver decided to switch the lane. This encouraged a vehicle approaching

from behind to increase its speed to take the empty space. While Switching, our user turned the LKA ON and took hands off the steering wheel, without knowing that the vehicle was in hesitation phase. This led to a tough turn to its old lane. Due to shock, none of the drivers could handle the case, which led to a collision between both automobiles.

Cyber-Security Overlap

The introduced scenario could also occur due to a Cyber attack. where an attacker is able to take control over an ECU and send an illegitimate LKA frame in critical time. This could lead to an equivalent damage. If this behavior was due to a Cyber-attack, some approaches would have been able to detect the abnormal behavior. This will be discussed in detail in the following section.

Proposed System Specification

Generally the required solution needs to conclude the relation between LKA and RLD frames, as well as how a change in one of them could affect the other. The model needs to understand the behavior of safe driving and detect unsafe behavior, such models must utilize context based analysis.

The proposed solution must be able to detect unsafe exchange of control between the user and the machine. It also must be able to deal with various RLD patterns. This includes noisy or distorted RLD values. Subsequently the model must deal with the sudden environmental change as well as the sensor performance limitation.

At last, the solution must respect the constraints introduced by the automotive industry. On top of those constraints the model size and inference speed in real time fashion.

Related Work

In [10] they worked on the formation of the SOTIF standard, while [11] was concerned with its development and the extension of the standard to multiple vehicles. Other approaches such as [6] were concerned with identifying the challenges that could meet the testing process. Proposal of new verification and validation methods were the subject of [5, 7]. Special concern was brought to the safety of neural networks in [12, 13]. [3] discuss different strategies to achieve safety in engineering, while [14] studied the confidence between the driver and the automated driving system. On the other hand few papers discuss specific practical cases, or provide a certain case study to work on such in [15].

As hinted in the Cyber-security overlap section, some models that are developed for Cyber-security purposes could be used in the safety context. In this section we discuss some of these methods.

At first, signature based methods. Where an unsafe scenario could be identified through a unique pattern. Such methodology will not be convenient since the number of

unsafe cases and scenarios are countless. There will be new scenarios that occur in real life that the vehicle never passed by in any simulation. So despite the high mitigation ability of signature based solutions for already known unsafe scenarios, these methods will be unable to solve new ones.

Secondly, the Finger-Printing techniques which try to differentiate between different ECUs based on their physical characteristics. In [16] the authors proposed a clock based solution while in [17] they used voltage measurements.

These techniques may be able to solve such scenarios if it was caused by a Cyber-attack. Where the attacker gained control over a random ECU and impersonated the Steering Column ECU. While in our case, no impersonation occurs, the legitimate ECU (Steering Column ECU) is sending a legitimate frame (LKA) where the user is overriding the lane keep assistant. These all are characteristics of normal cases where no change occurs in terms of the physical properties of ECU.

At Last the machine learning based solutions which can be a good candidate for solving such problems. Where a machine learning model could be trained to understand a normal safe behavior. So that it can be able to deal with cases that it has never dealt with before. Such models were used in [18]. However, ML based solutions may introduce safety threats.

Proposed Model

In our proposed work, we rely on ML based techniques to understand the pattern between inputs and deal with complex new scenarios. On the other hand we mitigate the safety threats introduced by ML techniques by following the SOTIF standard guidelines through design, verification and validation.

As discussed in the Safety Critical Scenario section, the nature of the introduced scenario is unique. It introduces a misuse of a legitimate function which causes damage to vehicles without any electrical or electronic failure. Such scenario could not be observed by checking normality of frames at a point level, therefore a context based model that is capable of realizing the relation of multiple inputs and their impact on one another is required. For this task we propose Predictive Models.

In order to meet specification listed in the Proposed System Specification section, we decided to design a ML model that is concerned with evaluation of LKA transitions (triggering event). Inference time and model size will not be points of optimization in this work. We will suffice with acceptable values for them.

Predictive Models

They are the models that use current data to predict the future or following state. Such models could have the ability to understand the dependencies of an input on other inputs. We propose two models: the first is LKA Predictor, the second is the enhanced version of LKA Predictor. We compare the performance of both models with the baseline model. The

baseline model is a non contextual based model. We use this comparison to study the importance of contextual analysis for this task.

Baseline

Our baseline consists of Three components. The first is the Filter block which is responsible for extracting the RLD and LKA frames from data flow. Secondly the Memory Unit which acts as a storage unit. At last, the Predictor is the unit which gathers all information to make decisions about the current LKA state.

As shown in Figure 4, we use the simple linear layer that has two inputs. RLD value is the first input, while previous LKA state is the second input. The objective of the model is to predict the next LKA state.

LKA Predictor

LKA predictors consist of four components. Filter, Memory Unit and Predictor, which are used in the same way as in the baseline. In addition, the History block, which is responsible for creating a compressed representation for all its inputs.

As Illustrated in Figure 5, the Filter receives the data flow and filters out the LKA and RLD frames using their IDs, both signal are forwarded to memory units to be stored, history is carrying a representation for all past frames of RLD frames,

FIGURE 4 LKA predictor baseline architecture, the model filters RLD and LKA value from data flow, the memory unit operates as a delay unit, RLD value and previous value of LKA are fed to the **predictor unit** to predict the next LKA state

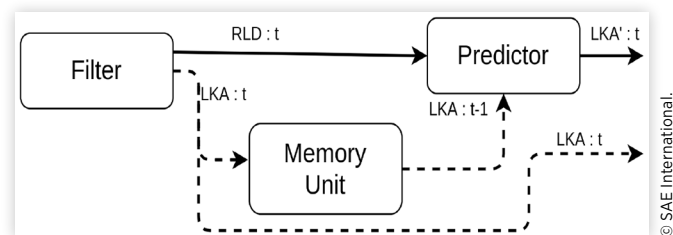
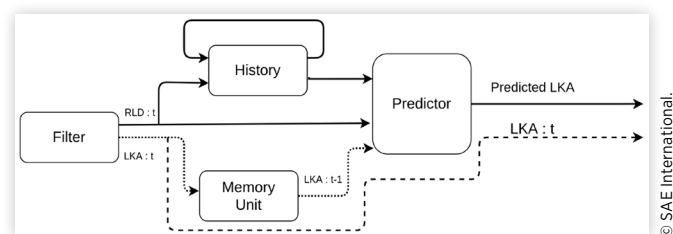


FIGURE 5 LKA predictor architecture, the model filters RLD and LKA value from data flow, the memory unit operates as a delay unit, the history unit creates a condensed representation of RLD values. History representation, RLD value and previous value of LKA are fed to predictor unit to predict the next LKA state



Predictor is the decision maker which receives the current, history of RLD, and the previous LKA state.

LKA Predictor Components

- 1) Filter: In addition to the selection of RLD and LKA Frames, it is responsible for modifying the values of RLD. We modify the values of RLD when we use augmented data to add noise.
- 2) History: Is convolution layers that stacks its existing representation with the current input frame and produces an output of the same size as input (represent the full sequence of frames in size of one frame).
- 3) Predictor: Consists of 3 Linear layers that increase the features to 1024 then a hidden layer that also outputs 1024 features and the last linear layer compresses features to LKA frame size. This represents the model prediction.

Detailed Operation

Algorithm 1 LKA predictor Algorithm

```

lka_id  $\leftarrow$  LKAframe
RLD_id  $\leftarrow$  RLDframeID
lka_state  $\leftarrow$  zero(OFF)
RLDt  $\leftarrow$  normal RLD tensor of CAN frame size
histt  $\leftarrow$  zero
while minibatch in dataset do
  if minibatch has frame of ID of lka then
    | lkat  $\leftarrow$  xor(lka_id, 1)
  end
  if minibatch has frame of ID of RLD_id then
    | RLDt  $\leftarrow$  RLDframe
  end
  hist  $\leftarrow$  History(hist, RLDt)
  lkapredicted  $\leftarrow$  Predictor(lkat-1, RLDt, history)
  lkat-1  $\leftarrow$  lkat
end
loss = MSE( lkat, lkapredicted)

```

We encourage the model to realize the dependencies between LKA state and RLD values, such design could be understood as a question to the model. The question is given:

- Current distance from the lane.
- Representation for all previous RLD frames.
- Last LKA state.

Predict what should be the next LKA state. In our scenario the question would simply be what should be the LKA state given that the vehicle is switching the lanes and the LKA is already OFF? The normal answer would be “it should continue

in the OFF state”, However what we really have in our scenario is a transition in LKA state to ON state while switching lanes. Therefore we must have an anomaly.

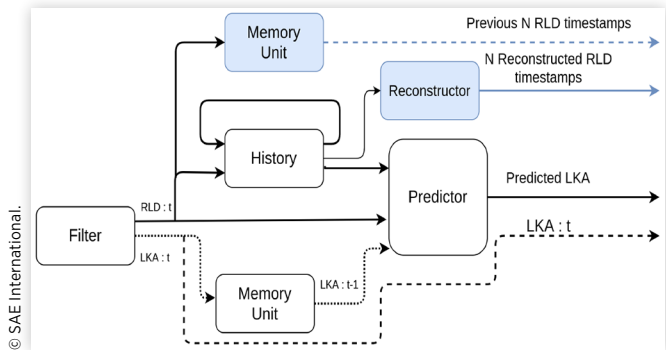
Enhanced LKA Predictor

Based on our results that will be discussed in the experiments section we decided to add an extra objective to the model to ensure that the History block is working as it is supposed to be.

In this version we added a reconstruction objective for the History block. In the first version the history block is trained on the objective of predicting LKA only. Such setting provides no guarantee that it is providing the Predictor block with historical representation of RLD. For example we may find that the function of the history block is just a replacement function where its representation carries only characteristics of the last RLD frame.

Our intuition about this new setting is that it guarantees that the History block is aiming to carry a representation for the last N inputs, which the block is responsible to reconstruct. Moreover, the hardness of reconstruction when the values of input are rapidly changing could be able to increase the loss if a transition occurred in this varying environment.

FIGURE 6 Enhanced LKA Predictor architecture, LKA predictor architecture, the model filters RLD and LKA value from data flow, the memory unit operates as a delay unit, the history unit creates a condensed representation of RLD values. The history, RLD value and previous value of LKA are fed to the predictor unit to predict the next LKA state. The history representation is fed into there constructor to reconstruct the RLD last (N) RLD values



Enhanced LKA Predictor Components

All components work similar to the first method. Except constructor block which is a trainable transpose convolutional layer with input channels equal to 1 and output channels equal to N. Where N is the number of previous instants to be reconstructed from the stored history representation.

It is clear that both versions have similar operations. Adding that new reconstruction objective to the model will enhance the quality of the model in some scenarios.

Algorithm 2 Enhanced LKA predictor Algorithm, Blue lines represent the differences with LKA Predictor

```

lka_id ← LKAframe
RLD_id ← RLDframeID
lka.state ← zero(OFF)
RLD ← normal RLD tensor of CAN frame size
histt ← zero
ws ← N (window size);
RLDlist ← empty list
while minibatch in dataset do
  if minibatch has frame of ID of lka_id then
    | lkat ← xor(lka_id, 1)
  end
  if minibatch has frame of ID of RLD then
    | RLDt ← RLDframe
  end
  hist ← History(hist, RLDt)
  lkapredicted ← Predictor(lkat-1, RLDt, history)
  lkat-1 ← lkat
  RLDreconstructed ← Reconstructor(hist)
  append(RLDt in RLDlist)
end
losslka = MSE(lkat, lkapredicted)
losshist = MSE(histt-ws:t, histreconstructed t-ws:t)
loss = 0.5 * losslka + 0.5 * losshist

```

Experiments

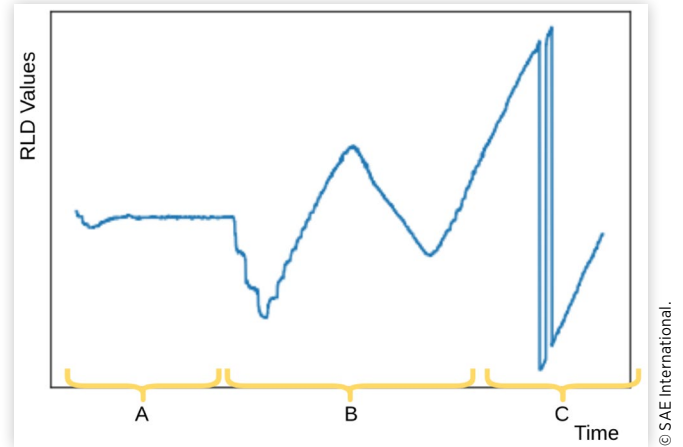
Training Details

Experiments are conducted on i5-8400 CPU 2.80GHz × 6 with graphics GeForce GTX 1070 and 16 GB RAM.

Dataset

- 1) Original Dataset: Our original dataset is recorded on a real vehicle. Normal behavior is recorded and a real safety-critical scenario is applied. The dataset contains the full flow of frames including LKA, RLD and the rest of frames as well. Figure 7 illustrates the values of RLD in the dataset. We can classify the values into three phases. Firstly, the constant phase denoted in (A) where the values have small deviations from their mean value. This corresponds to a vehicle moving in the middle of a lane. Phase (B) corresponds to the steering mode where a vehicle is about to leave the lane then its direction is modified to be centered in the lane. It consists of relatively higher and lower values of RLD than the RLD values exist in the constant phase. Last phase (C) is the switching mode where the vehicle has already left the lane. The hesitation in RLD values could be interpreted as the sensor is moving left away from the right lane, so it records high RLD values. As it passes over the new right lane which is of zero distance. The hesitation continued until the vehicle passed this phase and centered at the middle of the lane.

FIGURE 7 Original RLD dataset recorded from real vehicle scenario. It divides the picture into three categories. A represents Constant Phase, while B represents the steering phase, At last C is the switching phase.

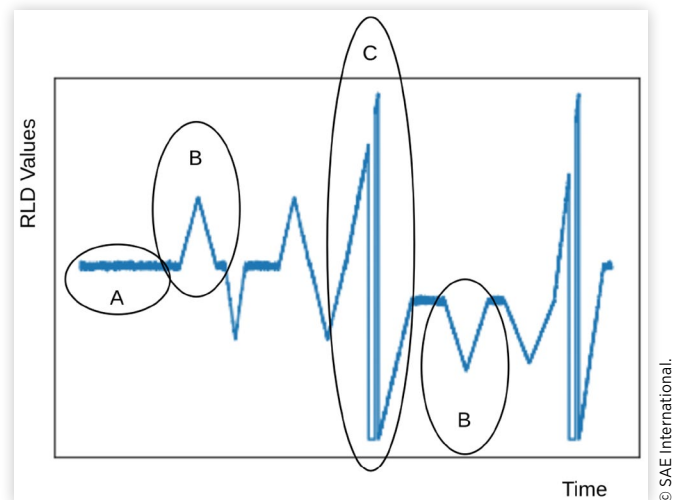


- 2) Augmented RLD: We decided to synthesize a dataset by using the distribution of the original dataset. This dataset allows enriching the model with various complex scenarios.

Using the phases explained in the Original Dataset we extract the distribution characteristics of the RLD and try to synthesize data with added noise of the same distribution.

We use Sine Wave with added noise to imitate the constant phase as in (A) in Figure 8. While we use the absolute value of x where x goes from $[-1, 1]$ to represent the steering phase as in (B). The switching or hesitation phase represented in (C) is a multiplication between a special setting of the absolute values of x and square waves.

FIGURE 8 Augmented RLD dataset. Part A consists of sine waves of small amplitude that represent the Constant phase. Part B is a representation of the steering phase using the graph of absolute x . At last Part C represents the switching phase which is graphed by multiplication of square wave and special setting of absolute x .



- 3) Augmented LKA: In order to avoid overfitting, we randomize the position of LKA transitions in each epoch. Synthesized LKA transitions are designed to occur only at the constant phase, thus the model can learn the safe timing of a transition.

Verification

Following the guidelines of the SOTIF standard, we list a set of test cases that directly targets the trigger event of the safety-critical scenario. The model should conclude that it is unsafe to change the LKA state while RLD is changing as in steering or switching phase. Following test cases are designed to test the safety of LKA transitions. Test cases use a clean RLD signal without any added distortion. Each case illustrated in [Figure 9](#) suggests a unique LKA-RLD scenario. Success in a scenario adds a conclusion about what the model understands.

In explanation of the following test cases, We use the statement ON for transition of LKA from zero to one.

Case 1: In this case LKA is OFF for the whole scenario with no transition. Best behavior must show no anomaly. Succeeding in this scenario means that the model understands the three phases of RLD. Also understands the ability of the driver to steer and switch lanes.

Case 2: A single legitimate ON transition. We expect the model to record an anomaly at the switching phase (very high or very low values of RLD). Since the vehicle cannot leave a lane while the LKA is ON.

Case 3: In this case we have four transitions. Three of them are legitimate transitions. all legitimate transitions exist in the constant phase. The illegitimate OFF transition occurs in the steering phase. If a model succeeds in such a scenario it means that the model understands that LKA transition can occur at constant phase and LKA transitions from ON to OFF can not occur at steering phase.

Case 4: It contains two transitions. A single legitimate ON transition and an illegitimate OFF transition. It ensures that the model realizes that LKA transitions from ON to OFF cannot occur at the steering phase.

Case 5: It has only illegitimate transition in the steering phase. Detecting such anomaly says that the model concluded that LKA transitions from OFF to ON cannot occur at the steering phase.

Case 6: Similar to case 5. However it ensures that the model understands that LKA transitions from OFF to ON can not occur at the steering phase at local minimum RLD.

Case 7: This case forms a train of transitions. The set of transitions which occur in the first phase is considered normal. Any transition else is considered an anomaly. We use this special case to determine the thresholds and calculate the precision and recall values. Note that we differentiate between the ON and OFF transitions since by experiments we found that the model gives different output for each transition even if both existed in the same position. This will be discussed in detail in the Determining Threshold section.

Validation

According to SOTIF, a model must be tested on unknown test cases. This process is called validation, it aims to ensure safety operation of the model in new scenarios. An exploration phase must take place to collect most of the cases that a model could pass by. In the verification phase we covered the safety of the LKA transitions. In this phase we concentrate on the distortion that could occur to the RLD signal. The introduced family of test cases clarify the importance of ML solutions, Due the countless illegitimate patterns of RLD that can not be covered by hard coded algorithms. In reality such cases could occur either due to sudden environmental change or due to limited performance of sensors.

[Figure 10](#) illustrates the shape of distortion added to the original RLD dataset. In this phase we fix the pattern of LKA. We use the pattern of LKA used in verification test case 7. We used test case 7 as it provides a rich LKA transition. This could help to better understand model behavior. To summarize, the final setting is train of transitions as LKA as in (g) in [Figure 9](#) and distorted RLD signals represented in [Figure 10](#).

In each case distortion is added to the same parts of RLD. We add distortion to two parts. The first part is in the constant phase the second part is in the steering phase. Each part includes 4 LKA transitions. Two as ON transitions and two for the OFF transitions.

In the first part we investigate how much distortion the model will tolerate before it classifies a safe transition as an anomaly. While in the second part we investigate if the model would classify an unsafe transition as safe due to added distortion.

FIGURE 9 Verification test cases. They represent a set of scenarios of safe and unsafe transitions. Examples represent scenarios where no transitions occurred, single safe transition, combination of safe and unsafe transitions and at last a train of transitions

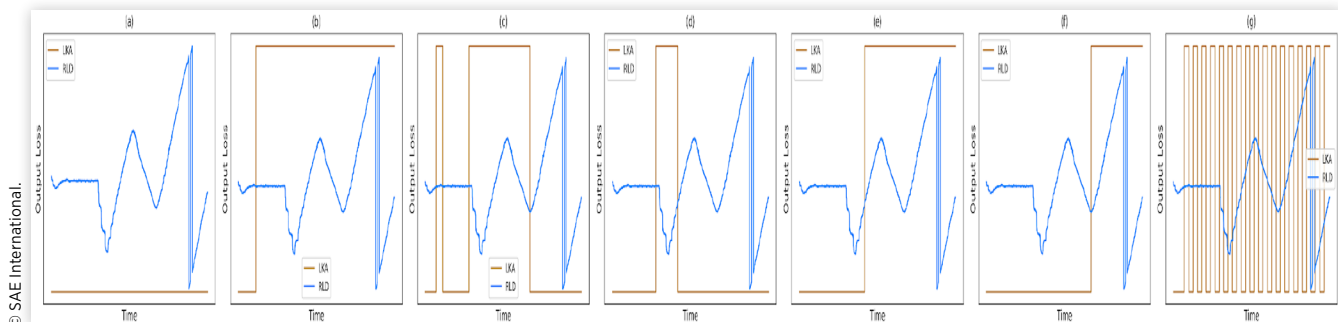
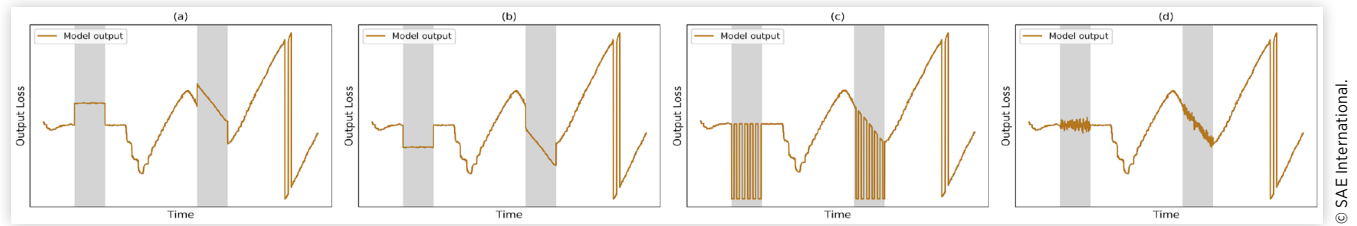


FIGURE 10 Validation test cases. They represent experiments concerned with anomalous patterns of RLD. They represent positive bias, negative bias, square wave and normally distributed noise. During training only we set the LKA transitions to be randomly positioned mostly in the constant phase. Such settings help the model to avoid overfitting. In addition we use the augmented dataset as the training set while the whole original dataset is used only as a validation set



In [Figure 10](#), each image represents a family of experiments. Positive bias **(a)**: This family tests the model against a positive bias. three different levels of bias are experimented. The bias values range from 0.1 to 1. In the low level experiment, we consider transitions occurred in the steering phase, as normal transitions. Because the RLD is kept in an acceptable range after distortion. Also the change in RLD value is considered small.

Negative bias **(b)**: Similar to positive bias, We use three ranges of negative bias. Image (b) represent the moderate negative bias level. We consider the low level experiment as normal for the same reasons discussed in the positive bias. Square signal **c**: In this case we use square signal distortion with high, moderate and low frequencies. Image (c) represents a moderate frequency experiment. We chose square waves due its similarity to the hesitation phase pattern.

0.08Hz, 0.04Hz, 0.02Hz are the high, moderate, low values of frequencies respectively. We found 0.08Hz square signal was able to diminish the original signal characteristics, so we used it as the high frequency case.

Normal distribution **(d)**: In the last type, we have three cases of experiments. We use the normal distribution with mean zero, while standard deviation is equal to the original dataset standard deviation at the steering phase for the low level experiment. In the second case the standard deviation of the noise is double the original dataset standard deviation, while the last case is ten times the original dataset standard deviation. Image **(d)** represents moderate level Gaussian noise.

We consider the transitions occurred in the constant phase of the first two cases as safe. Since the signal did not lose its characteristics after adding the distortion.

Models Configuration

In our experiments we use three models. **Baseline** we use the simple linear layer that has two inputs and single output. **LKA Predictor** which consists of convolution layers as History block and 3 Linear layers that increase feature size to 1024 and reduce it back to frame size. At last **Enhanced LKA predictor** which has the same blocks as LKA predictor with extra convolution layers for reconstruction.

For all models we use a similar training configuration. We use Adam Optimizers with betas equal to 0.9 and 0.999. We used a learning rate of $1e-4$ that was determined by grid search. Models are trained for 4 epochs with batch size of 8

and sequence length of 10. In addition to mean square error as a loss function.

We initialize the current and previous LKA state variable to zero which denotes the OFF state. The RLD state is initialized to a normal RLD frame with moderate value. Moderate values represent the constant phase. The initial value that is input to the history block (**Hist**) is initialized to zero. In the Predictor block we found that increasing the size of the linear layer, sharpens the transitions. which means that the model outputs sharp triggers at transitions as in 12. Therefore we set the hidden size of the linear layer to 1024.

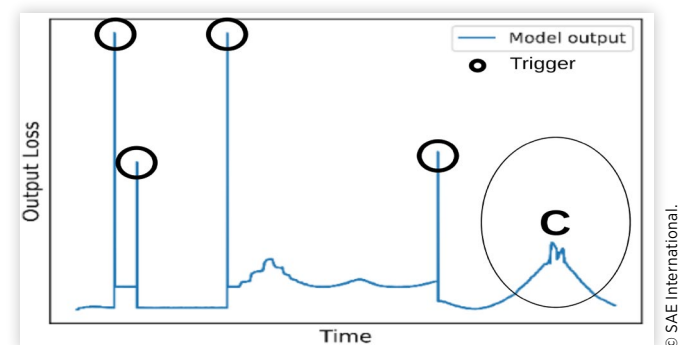
Determining Threshold

During evaluation we observed a special behavior that is repeatedly occurred by different models. The output loss for the ON and OFF transitions are usually different. Even if both were positioned in the same point in different scenarios or in similar phases of RLD.

As illustrated in [Figure 11](#) the initial ON transition recorded much higher loss than the one recorded by the following OFF transition. Despite the fact that the first 3 transitions are normal, there is a large difference between the loss of both types. Therefore, determining a single threshold that separates normal and anomalous data is impossible.

For this, we suggest multiple thresholds for each type of transition. For each model we use five thresholds. Each type

FIGURE 11 The output of enhanced model on the verification test case3, high triggers corresponds to ON transitions, As the first and third triggers. Short triggers correspond to OFF transitions



has an upper and lower bound. Those thresholds will be determined on the seventh verification test case. The last threshold is the one that covers the model output at the non LKA transition points. The setting is illustrated in 13.

The technique we used is illustrated in Figure 13. To set the bounds of the safe ON region we had three choices.

Firstly, the tight choice, to make the thresholds equal to the transitions that are on the tip of the safe region. Secondly, the wide choice, we can make the thresholds equal to the closest unsafe ON transitions. Thirdly, to take the average distance between the highest safe ON and the closest unsafe ON transition. We chose the third choice. to avoid high sensitivity of the model. Similarly we determine the Lower bound.

$$T_{Upper\ On} = \frac{B_{Least\ ON\ Anomaly} + B_{Upper\ ON\ Normal}}{2}$$

T stands for threshold and B stands for bounds.

Note that this method depends on our observation of the output pattern of the models. As shown in Figure 11 the output of the models at the LKA transition point is large, with low loss at non transition points. In the baseline model the output is different, it is a continuous overlapping pattern without large output at the transition point, therefore we determine thresholds manually for the baseline.

FIGURE 12 Represent safe and unsafe zones bounded by 5 thresholds. Upper and lower bounds for each transition type and non transition area bound. Dark areas are safe regions while white represent unsafe regions

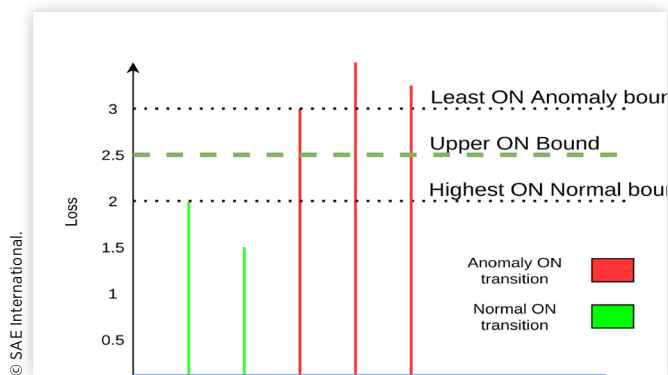
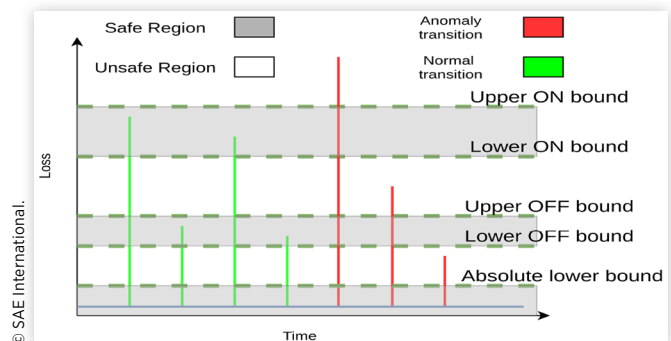


FIGURE 13 Represent ON transitions only. Green lines are safe transitions, while red are unsafe. Upper on bound are determined by averaging the highest normal value and the least unsafe trigger value



Verification Results

In Table 1 shows the result of the models upon each verification test case. F stands for true positive equal to zero, which makes the f1 score equal to infinity. LKA predictor and Enhanced LKA predictor models were able to pass all test cases. They were able to successfully classify the safety of transitions. In Figure 11 the area (C) stands for the output of the model at the switching phase while LKA is ON. In verification test case 1, 4 and 5 both models record higher loss for the same phase, which means that the model can conclude the safety of the LKA-RLD pairs. While in the case of baseline, there was an overlapping between safe and unsafe transitions. In addition, the baseline model had no sharp triggers for the OFF transition, which added complexity in determining the thresholds.

Despite the huge difference between parameter size among all models, There was no significant difference in inference time. Our conclusion is that the Filter block consumes most of the inference time. This makes no difference between models in terms of inference time.

Validation Results

Table 2 shows the f1 score calculated for each validation test case. The ability of both models are similar. However, LKA

TABLE 1 The table represents a comparison between baseline, LKA predictor and Enhanced LKA predictor. It shows the parameter size of models, mini-batches processed in seconds and f1 score obtained on verification test cases

Model	Parameter size	Inference speed mb/s	1	2	3	4	5	6	7
Baseline	2	139.1	0.8	0.53	F	0.49	0.6	0.42	0.8
LKA Pred.	1054726	138.1	1	1	1	1	1	1	1
Enhanced LKA Pred.	1054746	139.6	1	1	1	1	1	1	1

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TABLE 2 Table represents comparison between LKA predictor and Enhanced LKA predictor. It shows the f1 scores obtained on the validation test cases. Experiments 1-3 represent the positive bias family, 4-6 represent the negative bias, 7-9 show the square family, 10-12 represent the normal distribution experiments. Each family arranged from high to low ranges

Model	1	2	3	4	5	6	7	8	9	10	11	12
LKA pred.	1	1	0.8	1	1	0.8	1	0.98	0.96	0.98	0.95	0.88
Enhanced LKA pred.	0.98	0.96	0.78	1	1	0.8	1	0.98	0.98	0.96	0.95	1

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predictor has higher classification ability especially in the positive bias family. Both models get identical results in the negative bias family and square wave family, on the other hand Enhanced LKA predictor achieves slightly higher scores for the normal distribution family. Adding reconstruction objectives in the Enhanced LKA predictor came with harm on the positive bias family, slightly enhanced results for the normal distribution family and no effect for the others. Our intuition is that the new objective made the Enhanced predicted model more able to work in distortion without being affected with it. This feature is still important, since a super sensitive model will not be practically able to operate in real scenarios. Methods to tune this feature in a direction of research.

Discussion

Predictive Models are characterized by their instantaneous response to an anomaly. In addition to high precision recall values, they also can achieve similar results with augmented datasets and are able to realize LKA - RLD pair abnormality. On the other hand such models are highly sensitive to RLD variations. In addition to the complexity of setting the thresholds. At last such models do not always produce the same results with the same configurations which is a direction or research in the following work.

Despite the performance boost achieved by the shown ML models, we think ML models are not yet ready to act as a stand alone tool that provides sufficient safety. In order to produce a fully reliable solution we need an extra on-road testing phase, that includes a vast spectrum of users with different behavior and in different scenarios. In addition to this, Introduction of ML models for safety is not aiming to replace ISO 26262 that is concerned with electrical/electronic failure.

Summary

Increasing demand on autonomy in the automotive industry made the usage of AI models inevitable. However Such models introduce a serious threat to automobile safety and security. ISO/PAS 21448 SOTIF is a safety standard that is designed to deal with risks due to non electrical/electronic failure. Unlike most SOTIF-related papers, we aim to put SOTIF into practice, instead of discussing the standard. Subsequently, After giving an overview on the SOTIF standard, we introduce a conceivable safety critical scenario that targets the lane keep assist function. We use the suggested modelling technique in the SOTIF standard to analyze the scenario and extract the trigger event. Secondly, We discuss other studies in the safety domain and ML approaches introduced in similar domains. Then we propose a contextual based predictive ML model to monitor the intervention between the driver and lane keep assist system. Predictive models are those which use current data to predict the following/future data. We suggest the usage of predictive models for anomaly detection objectives. We also

studied the compatibility of the models with the constraints of an automobile. Thereafter, we singled out a section to explain our suggested methodology of determining thresholds. At last we follow the guidelines of SOTIF through verification and validation processes, where we study the behavior of the model against various scenarios and also against distorted inputs. For this we use a real safety critical scenario dataset as well as augmented dataset. Our experiments show high precision/recall values that exceed 90%. It also showed that there is a trade-off relation between the precision/recall values and the sensitivity of models to some of its inputs.

References

1. Kirovskii, O.M., and Gorelov, V.A., "Driver Assistance Systems: Analysis, Tests and the Safety Case. ISO 26262 and ISO PAS 21448," *IOP Conference Series: Materials Science and Engineering* 534:012019, Jun 2019.
2. Gharib, M., Lollini, P., Botta, M., Amparore, E., Donatelli, S., and Bondavalli, A., "On the Safety of Automotive Systems Incorporating Machine Learning Based Components: A Position Paper," in *2018 48th Annual IEEE/IFIP International Conference on Dependable Systems and Networks Workshops (DSN-W)*, pp. 271-274, 2018.
3. Varshney, K.R., "Engineering Safety in Machine Learning," in *2016 Information Theory and Applications Workshop (ITA)*, pp. 1-5, 2016.
4. Varshney, K.R. and Alemzadeh, H., "On the Safety of Machine Learning: Cyber-physical Systems, Decision Sciences, and Data Products," 2016.
5. Liu, F., and Yang, M., "Verification and Validation of Artificial Neural Network Models," . In: Zhang, S., Jarvis, R., editors. *AI 2005: Advances in Artificial Intelligence*, (Berlin, Heidelberg, Springer Berlin Heidelberg, 2005), 1041-1046.
6. Koopman, P., and Wagner, M., "Challenges in Autonomous Vehicle Testing and Validation," *SAE Int. J. Trans. Safety* 4:15-24, April 2016.
7. Schumann, J., Gupta, P., and Liu, Y., *Application of Neural Networks in High Assurance Systems: A Survey* (Berlin, Heidelberg: Springer Berlin Heidelberg, 2010), 1-19.
8. SAE, T., "Definitions for Terms Related to On-Road Motor Vehicle Automated Driving Systems-j 3016," Society of Automotive Engineers: On-Road Automated Vehicle Standards Committee; SAE Pub. Inc., Warrendale, PA, USA, 2013.
9. "Iso/pas 21448: 2019," Jun 2020.
10. Schnellbach, A., and Griessnig, G., "Development of the ISO 21448," . In: Walker, A., O'Connor, R.V., and Messnarz, R., editors. *Systems, Software and Services Process Improvement*, (Cham, Springer International Publishing, 2019), 585-593.
11. Schwalb, E., "Analysis of Safety of the Intended Use (sotif),"
12. Burton, S., Gauerhof, L., and Heinzemann, C., "Making the Case for Safety of Machine Learning in Highly Automated Driving," . In: Tonetta, S., Schoitsch, E., and Bitsch, F., editors. *Computer Safety, Reliability, and Security*, (Cham, Springer International Publishing, 2017), 5-16.

13. Bodenhausen, U., "Architecture and Independence Controller for Deep Learning in Safety Critical Applications," . In: Bargende, M., Reuss, H.-C., Wagner, A., and Wiedemann, J., editors. *19. Internationales Stuttgarter Symposium*, (Wiesbaden, Springer Fachmedien Wiesbaden, 2019), 1131-1142.
14. Guo, M., Shang, S., Haifeng, C., Zhang, K. et al., "Control Model of Automated Driving Systems Based on Sotif Evaluation," SAE Technical Paper [2020-01-1214](https://doi.org/10.4271/2020-01-1214), 2020. <https://doi.org/10.4271/2020-01-1214>.
15. Burton, S., Gauerhof, L., Sethy, B.B., Habli, I., and Hawkins, R., "Confidence Arguments for Evidence of Performance in Machine Learning for Highly Automated Driving Functions," . In: Romanovsky, A., Troubitsyna, E., Gashi, I., Schoitsch, E., and Bitsch, F., editors. *Computer Safety, Reliability, and Security*, (Cham, Springer International Publishing, 2019), 365-377.
16. Cho, K.-T. and Shin, K.G., "Fingerprinting Electronic Control Units for Vehicle Intrusion Detection," in *25th USENIX Security Symposium (USENIX Security 16)*, Austin, TX, pp. 911-927, USENIX Association, Aug. 2016.
17. Cho, K., and Shin, K.G., "Viden: Attacker Identification on In-Vehicle Networks," *CoRR* abs/1708.08414, 2017.
18. Hawkins, S., He, H., Williams, G., and Baxter, R., "Outlier Detection Using Replicator Neural Networks," 2454:113-123, 2002.